Activity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



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Part Reasoning

The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

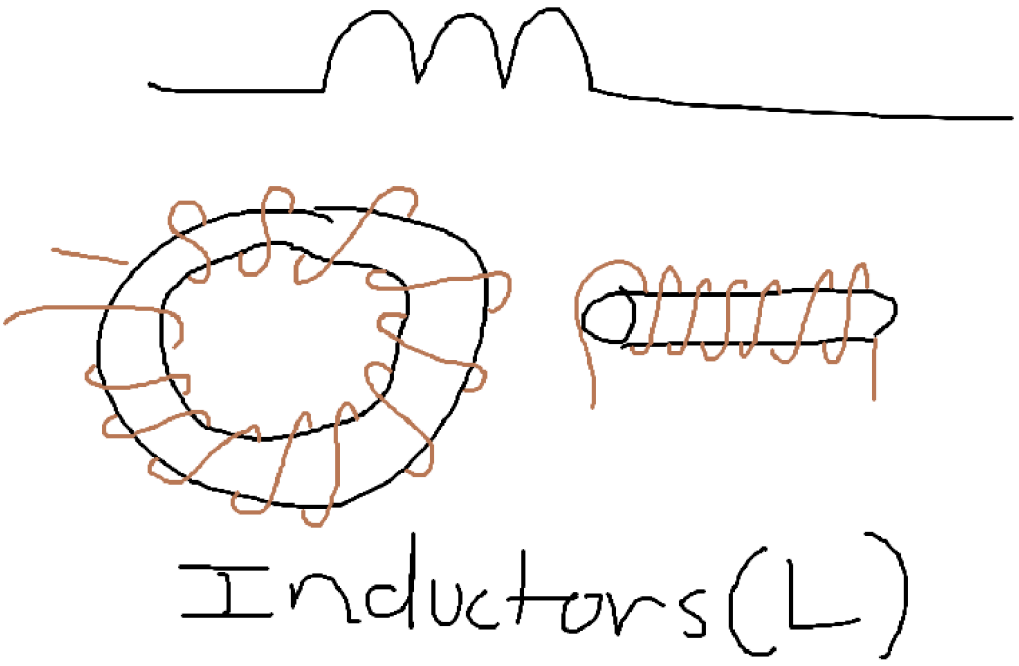
There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.
Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.

WHY THIS BUCK IC:

The BuckIC I chose is a 2A step-down converter usable at an input range of 4.5 V to 17 V, with an adjustable output. Pin 1 (GND) provides the ground reference for the IC on the low-side NFET. Pin 2 (SW) is the switching node that connects to the inductor and switches between the high-side NFET and low-side NFET. This is used to control current. Pin 3 (VIN) receives the input supply voltage, also the drain terminal for high-side power NFET. Pin 4 (VFB) is the converter feedback input. This regulates switching by detecting output voltage or current to remain stable. Pin 5 (EN) enables or disables the converter, with a high signal turning it on and a low signal turning it off. Pin 6 (VBST) provides a supply input needed for the NFET gate drive circuit. Connected to a 0.1-µF capacitor between VBST and SW pins.



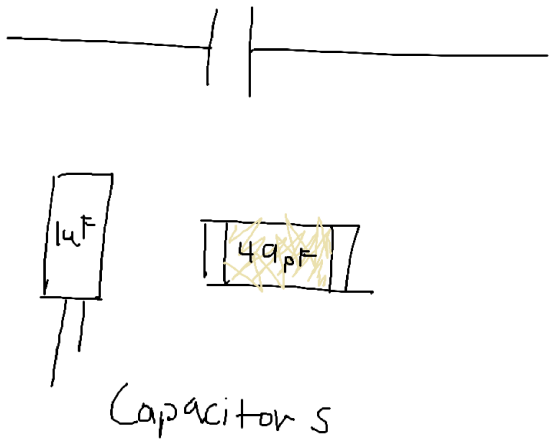
One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.

Why this inductor:

I selected the 4.7 µH inductor based on both the inductance value and the current ratings calculated based on the selected BuckIC. The datasheet recommends 4.7 µH for a 5 V output and specifically says the inductor's saturation rating must be greater than the calculated peak current, while its RMS current rating must be greater than the calculated RMS current. I calculated values of 1.22A peak-to-peak ripple current, 2.61 A peak current, and 2.03 A RMS current. Bourns SRN6045HA-4R7Y, which is a 4.7 µH inductor with a 7 A saturation current and

3.6 A rated current fits these specifications. The 7 A saturation rating is well above the 2.61 A peak current, so the inductor will not reach saturation during normal operation, and the 3.6 A current rating is above the 2.03 A RMS current, giving additional margin for heating. I chose this over an inductor with ratings closer to the minimum because of the lack of options closer to the actual specification, the cost, and also to have an additional margin. I chose this over other ones since we get a larger margin of error for a relatively low cost.

CALCULATIONS ON BOTTOM PAGES



Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.

Why these capacitors:

The way I picked my input capacitor was around minimizing the amount of input capacitors I would need and also cost. After looking at the 0603 options, I found a few options where using an 0603 capacitor would work for my project demands while staying within a reasonable amount of capacitors. This is because of the DC/Bias at 15V most of the capacitors would lose up to 80% of their efficiency and needing to hit 10µF, I would have had to use 10+ of them. Therefore I asked Edmund if I could switch to a 0805 capacitor which was approved. Using an 0805 capacitor I was then able to drop down to only 5 capacitors for a similar price. Below my selected parts is another table of the other parts I considered and why I didn't choose those

ones. Some of these are 0603 and some of these are 0805. This documents the iteration of my process. For the first output capacitor this was easier to find as since we only need them rated for the 5V output a lot fewer would be required and it would be easier to find them in the 0603 sizing. For the second output capacitor, I chose a 0.1µF capacitor in order to reduce output noise as mentioned above. I reused this 0.1µF capacitor as a bootstrap capacitor which was required in my BuckIC's documentation. When looking at the capacitors I decided which ones to pursue based on efficiency, size, and cost in that order resulting in my selected options.

Look at the table below to see 0805 capacitor.

The Secondary Table is on the page after the main one.

Resistor:

Why these resistors:

I need 2 resistors as in my BUCKIC documentation it requires 2 for the 4th pin VFB. I choose to have 1 resistor be 10 kΩ as a baseline as I was already given a formula in the BuckIC documentation. The other resistor was calculated to be 54.9 kΩ. This results in a voltage of 4.89V. While this isn't perfectly 5V it is safer than going up to the next resistor size and getting a voltage of 5.069 V as the specification is to be 5V and it is unclear if going over is safe. I have also provided an alternative resistor if going over the 5V limit to 5.069 V is safe. I based these resistors on cost, sizing, and the calculations to decide which ones to include.

See table below for alternative resistor.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.

COMPONENT TABLE:

Component	Cost (Individual x Units)	Link	Units	Size
Buck IC	$ 0.44000	buckic	1	
Inductor	$0.46000	inductor	1	0.236" L x 0.236" W (6.00mm x 6.00mm)
Input Capacitor	$1.70	input1	5	0.079" L x 0.049" W (2.00mm x 1.25mm)
Output Capacitor 1	$1.20	output1	4	0.063" L x 0.031" W (1.60mm x 0.80mm)
Output Capacitor 2	$0.13000	output2	1	0.063" L x 0.031" W (1.60mm x 0.80mm)
Capacitor Bootstrap (0.1µF)	$0.13000	bootstrap	1	0.063" L x 0.031" W (1.60mm x 0.80mm)
Output resistor 1	$0.10	resistor1	1	0.063" L x 0.031" W (1.60mm x 0.80mm)
Output resistor 2	$0.10	resistor 2	1	0.061" L x 0.031" W (1.55mm x 0.80mm)
Total Cost	$4.26			

Alternative Resistor 1	$0.10	alt resistor 1	1	0.063" L x 0.031" W (1.60mm x 0.80mm)

Note on Capacitor: This is an 0805 capacitor. Reasoning above.

Note on resistor: This is an alternative option to go above a 5V output. Reasoning above

PREVIOUS PART CHOICES:

[Why they don't work]

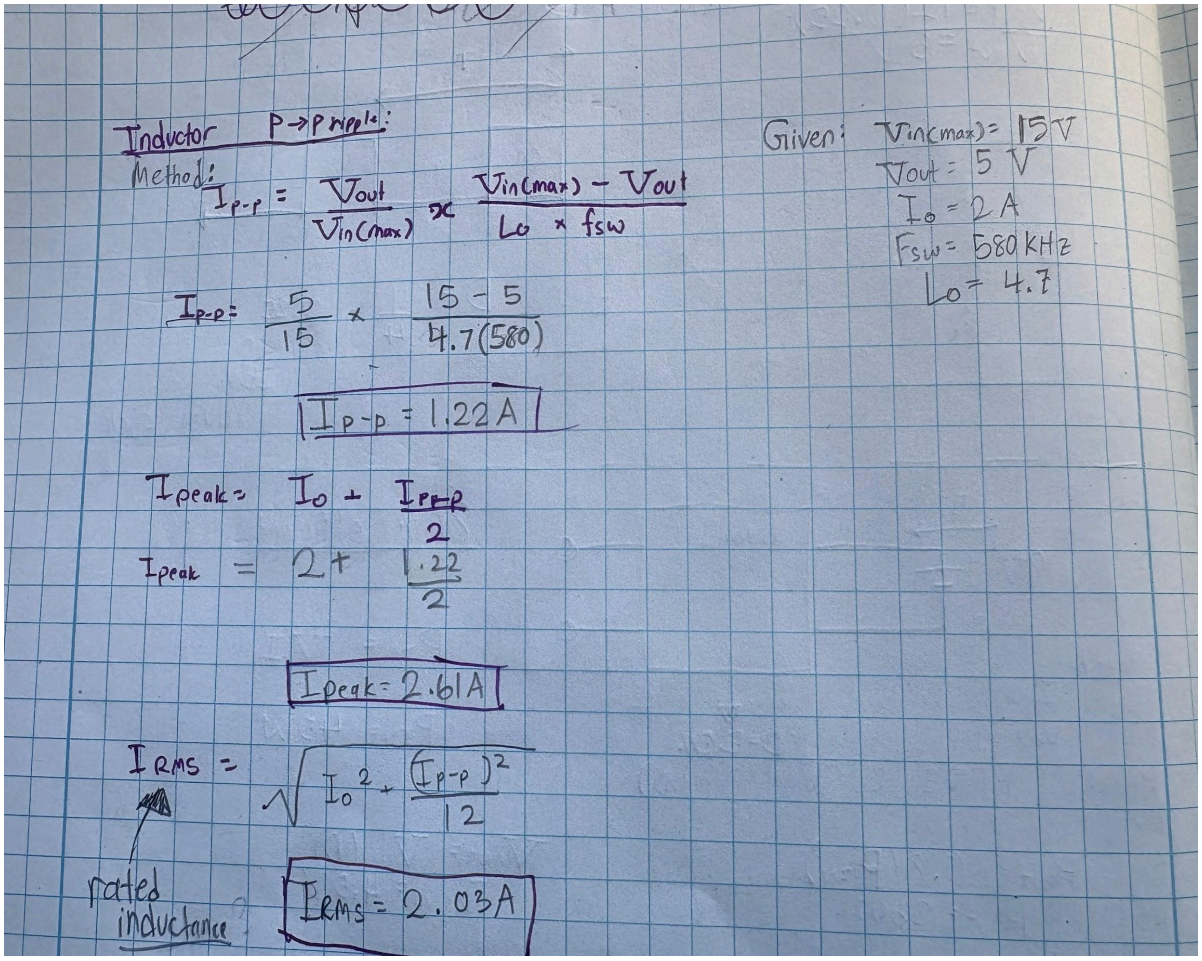
Part	Price	Link	Amount	Why it won't work
Capacitor	$0.63000	Link	3	This output capacitor won't work because of the sizing. The size needed for our project is 0603, this one is 1210. This was used as reference for pricing on output capacitors.
Input capacitor	$4.83000	link	1	This input capacitor is way too expensive, this was used as reference when looking at DC Bias charts to see if the capacitors I ended up picking would be good or not.
Output Capacitor	$1.09000	link	2	This output capacitor is pushing the price and also is too large. It is 1812 which is much larger than the 0603 specification.
Output Capacitor	$0.55	link	2	This was the recommended

				output capacitor from the BUCKIC documentation. I used this as reference when searching for other output capacitors, but it doesn't work since it is larger than 0603.
Capacitor Input	$0.29000	Link	4	This input capacitor is too large for the project.
Input Capacitor	$2.16	input capacitor	20 (only need 17 but buying 20 is cheaper)	Requires way too many capacitors to be effective
Input Capacitor	$1.50	input capacitor	10	Another solid option but requires too many input capacitors
Input Capacitor	$0.16700	input capacitor	10 (Buying 10 only need 6)	Decent option but still too many capacitors.

Specifications for project:
Currently use 0603, we don't typically need anything larger aside from that value. If we went above we would need a good reasoning. Output is for filtering. I'm using 22 micro henrys. That filters noise at specific frequencies, but I would need to filter noise at other frequencies. Spread out capacitor values so there is a wider range of frequencies being filtered.

Other Input Considerations:
GRM21BR61H475ME51L - 9 (Too Many 👎)
GRM21BC81H475KE11L - 6 (Price 👎)
GRM219R61C226ME15L - 5 (Price 👎)

Calculations



Inductor P→P ripple:

Method:

$$I_{p\text{-}p} = \frac{V_{out}}{V_{in(max)}} \times \frac{V_{in(max)} - V_{out}}{L_o \times f_{sw}}$$

$$I_{p\text{-}p} = \frac{5}{15} \times \frac{15 - 5}{4.7(580)}$$

$$\boxed{I_{p\text{-}p} = 1.22\,A}$$

$$I_{peak} = I_o + \frac{I_{p\text{-}p}}{2}$$

$$I_{peak} = 2 + \frac{1.22}{2}$$

$$\boxed{I_{peak} = 2.61\,A}$$

$$I_{RMS} = \sqrt{I_o^2 + \frac{(I_{p\text{-}p})^2}{12}}$$

$$\boxed{I_{RMS} = 2.03\,A}$$

rated inductance

Given:

$V_{in(max)} = 15\,V$

$V_{out} = 5\,V$

$I_o = 2\,A$

$F_{sw} = 580\,kHz$

$L_o = 4.7$

When looking for a inductor need a inductor with a rated current > 2.03A ??
greater than 2.61A

Inductance:

$L = \frac{V}{(dI/dt)}$

$\frac{10V}{\left(\frac{2.61A}{(5/15)} \cdot 580\right)}$

$dV = V_{in} - V_{out} = 10V$
$dI = I_{peak} = 2.61A$
$dt = D \times t_{sw}$
$D = V_{out}/V_{in}$
$t_{sw} = 580\,kHz$

$V_{in} = 15$
$V_{out} = 5$
$f_{sw} = 580$
$D = 5/15$
$\Delta I_{max} = 2.61$
$\Delta I_{rip} = 1.224$

$L = \frac{(V_{in} - V_{out}) \times D}{f_{sw} \cdot \Delta I_L} =$

2.2019 uH ??

$$L = \frac{V_{out} \times (V_{in(max)} - V_{out})}{V_{in(max)} \times f_{sw} \times \Delta I_{ripple}}$$

$$= \frac{5 \times (15 - 5)}{15\,(580{,}000)(1.22)} = 4.71\ \mu H$$

Capacitor Selection

$$I_{CO(RMS)} = \frac{V_{out}(V_{in} - V_{out})}{\sqrt{12} \times V_{in} \times L_O \times f_{sw}}$$

$$= \frac{5(15-5)}{\sqrt{12}(15)(4.7\mu H)(580\,kHz)}$$

$$I_{CO(RMS)} = 0.35A$$

The capacitor needs to handle a 0.35A of RMS ripple current.

The datasheet requires 4 capacitors as stated on page 14:

- 1 input
- 2 ouput
- 1 bootstrap

~~can be the same~~ 2?

better to choose seperate...

input:

- ceramic
- $\geq 10\mu F$

output:

- $20\mu F \leq x \leq 68\mu F$
- Ceramic
- Low-ESR capacitor..

For Resistor

From Buck Datasheet

$$V_{out} = 0.768\left(1 + \frac{R_1}{R_2}\right)$$

$V_{out} = 5V$

Take $R_2 = 10k\Omega$ as standard...

$$5 = 0.768\left(1 + \frac{R_1}{10}\right)$$

$$5 \div 0.768 = 1 + \frac{R_1}{10}$$

$$6.5104 - 1 = \frac{R_1}{10}$$

$$R_1 = 55.1k\Omega$$

Closest one is a $54.9k\Omega$ resistor to keep below 5V...

PCB and Schematic Design

Schematic:

When making my schematic, one of the problems I ran into was finding a symbol for my TPS562208DDCR buck converter IC. The symbols I found online were way too large for my schematic, so after talking to Edmund, I decided to make my own using the pin information from the datasheet. I also had some trouble with the wiring, especially connecting the inductor and output capacitors to VOUT. Even though the wires looked connected, KiCad wasn't always recognizing them correctly, which caused some of the wrong nets to show up when I transferred everything to the PCB. I had to go back and check the wires and labels several times before getting everything connected correctly.

PCB:

When making my PCB, I ran into several problems with the footprints, wiring, and component placement. One of the biggest issues was getting the footprint for my buck IC to match the datasheet, so I had to manually adjust the pad sizes and positions. I also had trouble with the VOUT connection because my inductor was originally showing GND instead of VOUT. I had to go back to my schematic and fix the connections before updating the PCB. After finishing the routing, I ran the Design Rules Checker and got errors about the courtyard around my buck IC and the missing board outline. The courtyard problem took a while to figure out because the issue was a minor error on the end location of the segments. After going through these problems, I was eventually able to get zero errors, warnings, and unconnected items.